

Haptic Glove With MR Brakes for Virtual Reality

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Abstract—Haptic gloves open up the world of force feedback by allowing the user to pick up and feel virtual objects in a natural way. In most of the existing gloves, a remote box houses a large number of actuators and sensors. Power to the glove is transmitted via cables. If the haptic gloves were smaller, lighter, and easier to use and control, they could become more common as human-machine interfaces. Recent developments show that actuators based on active fluids, such as the magnetorheological (MR) fluids, can be viable alternatives in haptics. But these devices are desk- or floor-mounted and use relatively large MR brakes. In this research, we developed a compact MR brake that is about 25 mm in diameter, weighs 84 g, and can apply up to 899 N-mm torque. The compact size was achieved by stacking steel and aluminum rings to create a serpentine flux path through the fluid. Six brakes were used to build a force feedback glove called MR glove. The glove weighs 640 g and does not require any remote actuators. Results of usability experiments showed that the MR glove improved task completion times in grasping virtual objects and could convey stiffness information to the user.

Index Terms—Force feedback, haptics, magnetorheological fluid (MRF), virtual reality (VR).

I. INTRODUCTION

VIRTUAL reality (VR) is a form of human-computer interaction where the user feels immersed in a simulated world through the rich and real-time sensorial interaction provided by stereographics, 3-D sound, and tactile feedback. Haptic feedback in VR is a term used to refer to touch- and force-feedback modalities. Force feedback opposes the user's motion and is intended to convey information on virtual object hardness, weight, and inertia.

Traditional applications of VR range from military simulations to arts and education, and from medicine to manufacturing. When haptic feedback is incorporated into the VR simulation, it increases the realism of the simulation and expands the application domain of the VR technology.

Most force feedback devices rely on electric motors, pneumatics, or some other conventional power producing method. Recent developments show that actuators based on active fluids, such as the electrorheological (ER) and magnetorheological

(MR) fluids, can be viable alternatives in haptics. MR fluids (MRFs) typically contain soft ferromagnetic powder suspended in silicone oil. When a magnetic field is applied across the volume of the fluid, chains of iron particles form in the fluid, thus inducing yield stress. A local shear stress larger than this yield stress must be applied to break the chains and allow flow or motion.

Many haptic devices using MR or ER fluids have been developed. They are floor- or desk-mounted. A floor-mounted rehabilitation device using linear MR damper was developed [1]. The device can be used to strengthen different muscle groups. It is a small, low-cost device that can be used at home. A desk-mounted parallel-link-type force display system using ER fluids allows motion in 2-D space and can present up to 50 N force back to the user [2]. Haptic black box (HBB-II) concept uses a cylindrical plastic container with MRF and a series of ferromagnetic cores, positioned around the container and used to dynamically address the magnetic flux inside the fluid [3]. The user can insert his/her hand into a latex glove in the device and interact with the MRF perceiving and discriminating different virtual objects. An MR joystick was constructed of two disc-shaped MR actuators positioned perpendicularly with a gimbal structure that transfers the movement of the joystick handle into two actuator rotary movements [4]. The device was used in a virtual maze game and virtual surgery application. An MR damper was coupled to a steering wheel to provide force feedback to the driver [5]. The wheel drives a virtual car providing force feedback in various simulated driving conditions. A training system was developed using an MR brake to reproduce stiffness of elbow joint of a patient to train a physical therapist [6]. A knee brace using ER fluid damper was designed to correct knee hyperextension during stance and stiff-legged gait [7]. Modeling and control of a five-bar haptic display was investigated using a commercial MR brake [8]. Results indicated that a first-order model can accurately predict braking torque for steps in applied current and is satisfactory for other types of inputs. The MR brakes eliminated many of the problems associated with friction brakes, including slow response time, stiction, and vibration. All of these devices are desk- or floor-mounted and use relatively large MR brakes.

An important interface is the haptic glove. It opens up the world of force feedback by allowing the user to pick up and feel virtual objects in a much more natural way. One of the first examples of a glove for VR was developed at the University of Tsukuba [9]. This was a string-based glove providing up to 7-N feedback force to the index finger and the thumb. The motors were placed on the dorsal side of the hand. It weighed 0.25 kg and provided force feedback by pulling the fingertips back as the user grasped a virtual object. Another string-based glove was the Laboratoire de Robotique de Paris (LRP) hand master [10]. This design provided up to 14 N force feedback to all fingers. The

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exoskeleton transmitted forces from 14 motors placed in a remote box via microcables, pulleys, and flexible links. In 1997, the Sensor Glove II was developed at the University of Tokyo [11]. The glove had 20 DOF, with each joint driven by motors through wire transmission to reduce the weight. The CyberGrasp was first introduced in 1998 and is the only commercially available force feedback glove [12]. An exoskeleton, weighing about 450 g, is mounted on the back of the hand to guide tendons to the fingertips. The glove can provide up to 12 N force to the fingertips. A specialized PC and umbilical device are used to power and control the CyberGrasp. The additional equipment makes transportation difficult since the control unit weighs 20 kg. The CyberGrasp has no position or flexion sensors, and therefore, requires an additional device, such as the recommended CyberGlove, to provide sensing. The Rutgers Master II—New Design (RMII-ND) was developed at the Rutgers Human Machine Interface Laboratory [13]. It uses a graphite-on-glass pneumatic piston for each finger, thereby significantly reducing the static forces when the glove is not powered. Pistons are directly attached to the fingers and are located between the palm and the fingertip, providing up to 16 N force to each fingertip. The working portion of the device is light and comfortable to wear, weighing only about 100 g. Each finger is equipped with sensors for measuring finger position, flexion, and abduction, along with displacement of the piston. The glove has a control unit that contains pneumatic valves, pressure sensors, and all the electronics. In addition, compressed air is required. Having the pistons in the palm limits the user's finger movements. A multifingered force feedback glove was designed using clutches and springs [14]. The glove employs a parallelogram linkage attached to the fingertips. The motion of the linkage is locked or unlocked by the clutches. The forces applied to the user's fingertips are a result of the deflection in the links and the springs at two of the joints at each finger mechanism. The maximum applied force is 3 N based on the holding torque of the clutches. A passive force display glove (PFDG) was developed using electromagnetic ON/OFF brakes [15]. Wire-pulley systems transmit braking force to the fingertips. A torsion spring is used to maintain tension on the wires. The four brakes are attached to the back of the hand. The weight of the glove is supported by a table-mounted robot. An innovative glove design uses the "encounter-type" force feedback [16]. The glove consists of finger mechanisms that follow the free motion of the user's fingers with the help of photodetectors. If the virtual hand is not in contact with a virtual object, the fingers of the glove follow the user's fingers with no contact. When the virtual hand makes contact with the object, the glove fingers make contact with the user's fingers and start applying forces. The design uses a circuitous joint that extends the link lengths in proportion to the joint displacement. The glove uses one motor on each finger and wire transmissions coupling three joints of each finger.

In almost all of these devices, a remote box housed a large number of actuators and sensors. Power to the glove was transmitted via cables. In the case of the RMII-ND, the cylinders were in the palm of the glove, but it had a large control unit with valves, sensors, and electronics. The PFDG glove used ON/OFF brakes mounted on the back of the hand but the weight

of the glove had to be sustained by a robot attached to the glove.

In haptic glove applications, the challenge is to design an actuator that is small enough to be placed on the hand yet powerful enough to restrict or stop the motion of the fingers as the user grasps a virtual object. The MR devices used in the current haptics applications are large, and therefore, not suitable to be used in a glove application. Devices that are grounded on the user's body can provide increased freedom to make the experience more natural. One disadvantage is that the user has to support the weight of the device. Considerations of weight and expense have probably limited the development of portable force feedback devices. If the haptic gloves were smaller, lighter, and easier to use and control, they could become more common as human-machine interfaces.

The primary goal of this research was to develop a force feedback glove (MR glove) that used compact MR brakes placed on the back of fingers [17], [18]. To the best of our knowledge, this was the first use of MR brakes in a haptic glove. The MR brakes are an attractive option because they are inherently safe due to energy dissipation and provide quiet, quick response with simple control. The glove uses six brakes, actuates three fingers, uses four-bar mechanisms to connect the brakes to the digits of the fingers, and does not require any remote box with actuators or sensors. For this application, we developed a new MR brake. It is a rotational actuator about the size of a U.S. quarter. It uses a unique serpentine flux path to maximize torque in a compact volume.

In this paper, we present the design of the MR glove and experimental results. Section II describes design of the MR brakes, including design goals and a serpentine magnetic flux path concept for performance improvements. Section III presents torque and time constant experiments with the MR brakes. Section IV details the MR glove prototype. Results of two usability experiments with the glove are described in Section V. Conclusions and future research directions are provided in Section VI.

II. MR BRAKE

The literature contains many devices using MRFs in applications such as optical polishing, fluid clutches, automotive suspensions, and as active dampers in civil engineering applications. In general, these devices involve either disc-type or valve-type designs [19]–[26]. In valve-type designs, the fluid is pushed through a narrow channel where the magnetic field is applied to control the flow rate, and hence the applied force. Typically, these designs resemble a cylinder-piston assembly with the coil on the piston shaft. In disc-type designs, the fluid is in a narrow gap between a rotating disc and a fixed outer casing. The coil is positioned close to the outer edge of the disc. When the magnetic field is applied, the increased yield stress of the fluid creates a braking torque on the disc. The existing MR devices are relatively large.

The primary motivation for this research was the need for a portable haptic interface that could be used in VR simulations. In this application, the challenge is in designing an actuator that

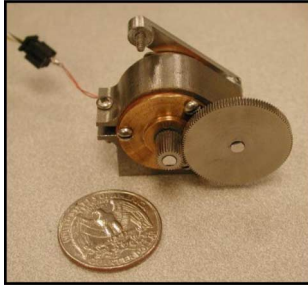


Fig. 1. MR brake.

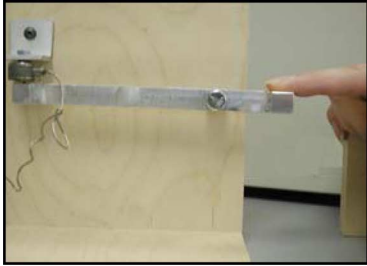


Fig. 2. Experimental setup to measure finger joint torques.

is small yet powerful. We used MRF to design a small passive actuator (brake) called MR brake (see Fig. 1).

The MRFs exhibit yield strength of 50–100 kPa for a magnetic field of 150–250 kA/m [27], [28]. Another such fluid, ER fluid, is controllable when subjected to an electric field. The ER fluids exhibit yield strength of 3–5 kPa in an electric field of 3–5 kV/mm. In this research, MRF was chosen because of its ability to exhibit higher dynamic yield strength. The fluid used in the brakes (MRF-240BS) was purchased from LORD Corporation [29]. This is a water-based solution that contains 84% solids by weight and can operate in the range from 0 °C to 70 °C. In the inactive state, the fluid has a viscosity similar to low-viscosity oil. Upon activation, it changes to a thick consistency similar to peanut butter.

A. Design Specifications

Our main design goal was to develop a brake that was no larger than 25 mm in diameter and no wider than the width of a finger (about 15 mm) yet powerful enough to apply sufficient force on a finger joint to restrict or stop its free motion.

We conducted an experiment with five male subjects to measure the metacarpophalangeal (MCP) (knuckle) and proximal interphalangeal (PIP) (medial) joint torques for the index and middle fingers. The subjects first rested their hand on a wedge with their index finger MCP joint on the edge of the wedge. Keeping their index finger horizontal, they pressed on a lever with their fingertip using their MCP joint. A load cell at the other end of the lever was used with a data acquisition system for data collection (see Fig. 2). Then, they repositioned their hand on the wedge such that their PIP joint was now on the edge of the wedge. The experiment was repeated to measure their PIP joint torque. The subjects were asked to apply as much force as they could without hurting themselves for a period of 5 min.

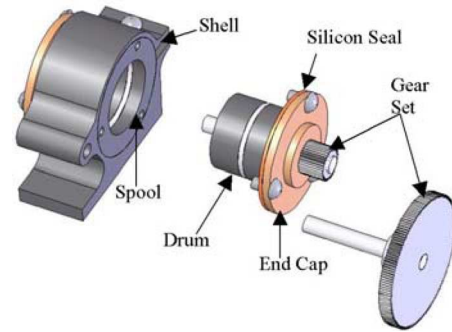


Fig. 3. MR brake assembly.

The goal was to measure the peak and sustained levels of torque by each joint. The subjects had at least 4 h of rest between tests.

The torque profiles between subjects were quite consistent. Initially, they applied a peak torque sustained for about 15 s. Then, the torque decayed sharply reaching a steady sustainable value in about 1.5 min. The average peak torques for five subjects' index MCP and PIP joints were 2300 and 900 N·mm, respectively. The average sustainable torques for the subjects were measured at 620 N·mm at the MCP joint and 220 N·mm at the PIP joint.

Typical object manipulation or grasping operations in VR do not involve peak torques. Instead, a user exerts slightly more force than necessary to grasp the object and sustains this force level. Furthermore, it has been suggested that force feedback gloves should be designed to specify a peak force capability of 20%–30% of the peak joint torques [30]. Therefore, the sustained torque values were used as the basis in determining the torque requirements for the brake design. Based on the results, 700 N·mm was estimated as the braking torque requirement for a large brake for the MCP joints. The braking torque requirement for a smaller brake for the PIP joints was estimated at 250 N·mm.

B. Design Details

The MR brake uses two concentric cylinders (see Fig. 3). The inner cylinder (the drum) rotates inside the outer cylinder (the spool). The gap between the drum and the spool is filled with the MRF that applies a controllable braking torque on the drum due to the change in its viscosity as a function of the strength of the applied magnetic field. The coil was wound on the spool and placed underneath the shell.

The torque transmissibility of the MR brake depends on the dynamic yield stress and viscosity of the MRF. Extensive dynamic modeling of MR brakes can be found in [31] and [32]. Often the behavior of controllable fluids is represented as a Bingham plastic having variable yield strength [21], [33]. The flow is governed by

$$\tau = \tau_y(H) + \eta \frac{\omega r}{h} \quad (1)$$

where τ is the total yield stress, η is the fluid viscosity, ω is the angular velocity, h is the fluid gap, and r is the radial position. The first term $\tau_y(H)$ in the equation is the dynamic yield stress

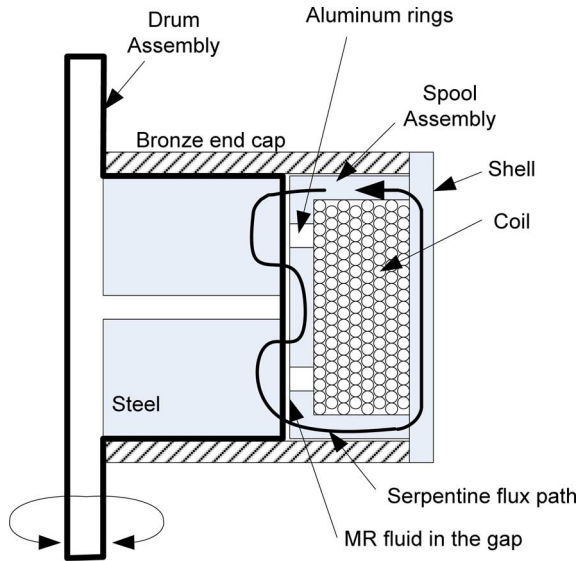


Fig. 4. Serpentine magnetic flux path. The figure shows cross-sectional view of the right half of a brake.

as a function of the magnetic field. The second term is the shear strain rate. During virtual object manipulation, the users move their fingers slowly due to inherent delays in the simulation and to be able to feel the interaction forces with the objects. The MR brakes rotate very slowly; hence, the viscosity term can be neglected. In practical applications, a small torque due to friction (e.g., due to the seal) also exists as a third component in the total braking torque. The significant portion of the braking torque is due to the dynamic yield stress acting on the outer surface of the drum. The total braking torque can be written as

$$T = 2\pi r^2 w \tau_y(H) + T_{\text{friction}} \quad (2)$$

where w and r are the width and radius of the drum, respectively. In conventional MR devices, flow channels are straight, and electromagnetic coils are configured so that flow streamlines are normal to magnetic field [19]. The magnetic coil is usually immersed in the fluid. The flux lines are shaped like a circle crossing the fluid at one or two places.

To increase the braking torque, as much of the MRF in the gap as possible must be exposed to the flux lines. The challenge is to fit a coil in the brake while keeping the physical size small and maximizing the flux lines crossing the fluid in the gap.

We developed a design with a serpentine flux path weaving through the MRF in the gap (see Fig. 4). While keeping the physical size constant, this design significantly improved the torque output compared to our initial designs with a single crossing. The flux path was shaped by using aluminum rings as insulators and steel rings as conductors of magnetic flux. The outer shell completed the flux path. The number of crossings is limited by the complexity of manufacturing the small insulating rings of aluminum that are 1 mm in width. With the capabilities of our manufacturing facility, we were able to build a 13-mm-wide prototype large brake with four crossings and an 8-mm-wide small brake with two crossings.

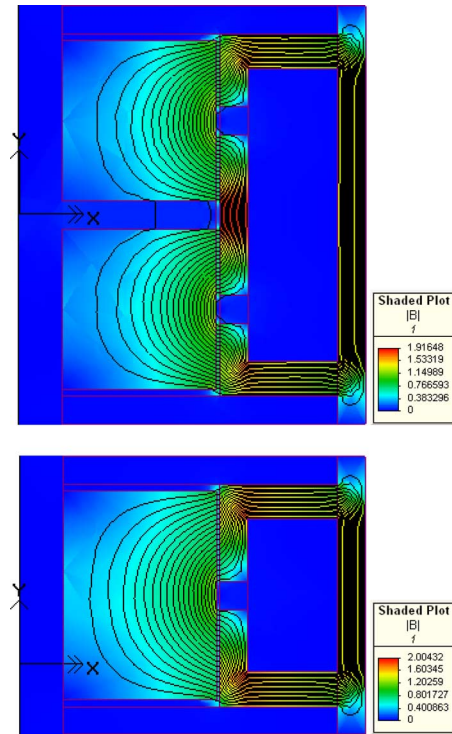


Fig. 5. Magnetic flux. (Top) Large brake. (Bottom) Small brake.

Due to the complex geometry, we modeled various brake designs using the MagNet finite-element method (FEM) software by Infolytica Corporation [34]. All material and magnetic properties, including the $B-H$ curve of the MRF, were entered into the software. The designs were optimized based on the magnetic flux density computed in the fluid gap. The goal was to maximize the flux density in the fluid, and hence the braking torque, while keeping the outer diameter of the brake about 25 mm.

The shear stress generated by the MRF is proportional to the magnetic flux through the fluid. If the number of coil turns and the current are increased, the flux will increase. However, this requires thicker wires and results in a larger coil. Using more turns of a thinner wire is possible, but this time, the wires overheat due to the increased current. The braking torque is also a function of the fluid gap, the drum radius, and width. The smaller the gap, the larger the magnetic flux in the gap since the relative permeability of the MRF is much smaller than that of low-carbon steel. Increasing the drum radius provides larger torque arm as the shear force is applied on the surface of the drum by the fluid. Furthermore, if the width of the drum can be increased, the total surface area where the shear stress is applied will increase. Consequently, to increase the braking torque, the fluid gap must be minimized, while the number of coil turns, current, drum radius, and width must be maximized.

After much iteration, the optimal design had a 7.125-mm drum radius and 0.125-mm fluid gap, producing 0.75 and 0.66 T flux density in the large and small brake designs, respectively (see Fig. 5). Based on the manufacturer's specifications for the MRF-240BS fluid, the shear stress corresponding to the

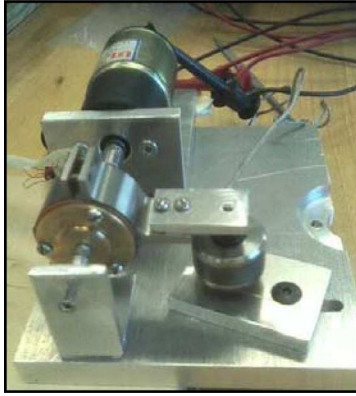


Fig. 6. Experimental setup for torque and time constant measurements.

aforementioned flux densities were 55 and 50 kPa. With these dimensions and the shear stress values, the braking torque output for an input of 0.5 A of current was calculated as 230 N·mm for the large brake and 130 N·mm for the small brake. The design also incorporates a gear train with 28:120 ratio resulting in computed torque output of 990 and 556 N·mm for the large and small brakes, respectively. These outputs exceeded the required torque specifications of 700 and 250 N·mm.

Both the spool and the drum contain rings made of low-carbon 12L14 steel with high magnetic permeability. They also contain thin rings of aluminum that act as insulators to the magnetic path. The magnetic coil was formed by wrapping 30-gauge magnet wire around the spool. The large and small brakes were wound with approximately 250 and 150 turns, respectively. Seals stamped out of silicon material were used to prevent the MRF from leaking out. The volume of MRF in the brake was about 0.1 mL. The end caps were made from oil-impregnated bronze to support and align the drum shaft concentric with the spool. In addition, the impregnated bronze provided a self-lubricated bushing to decrease the resistance between the end cap and the drum shaft.

III. MR BRAKE EXPERIMENTS

The torque output and time constant were measured using a test setup with a load cell, a lever arm, a dc motor, and a data acquisition system (see Fig. 6).

The braking torque experiment started with the dc motor freely spinning the brake drum. Then, the brake was engaged by applying 0.5 A current to the coil and the load cell voltage was recorded. The motor was sized so that the brake could not stall it even when it was fully engaged. Using load cell calibration data, the voltage was converted into the braking torque (see Fig. 7). The corresponding outputs for the large and small brakes were measured as 189 and 93 N·mm, respectively. With the 28:120 gear ratio, the torque levels were increased to 821 and 399 N·mm, respectively. When the brake was not activated, there was a slight frictional torque mainly due to the friction in the bushings, seals, and the MRF. The friction torque was measured as approximately 5 N·mm for both types of brakes including the gear train.

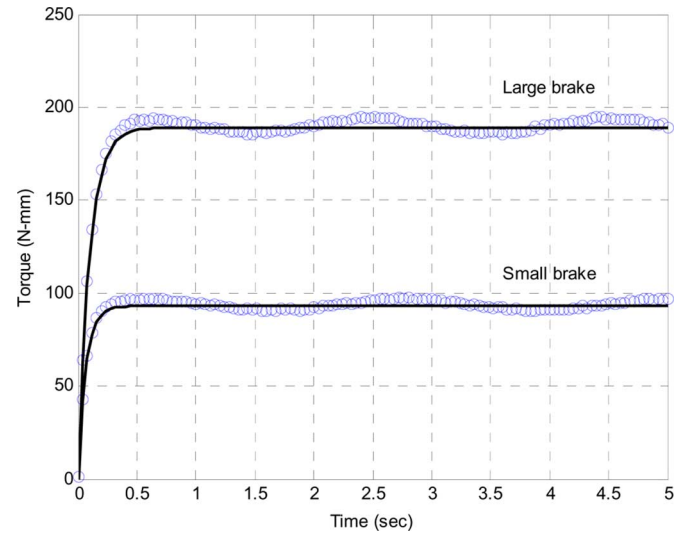


Fig. 7. Time response of the brakes. Straight line: simulated first-order response overlaid on experimental data.

The torque has a slight repeated pattern of fluctuation. Possible reasons for this are: 1) slight misalignment between the motor and drum shafts causing the load cell to wobble on its axis; 2) nonuniform distribution of the MRF in the small gap between the drum and the spool; or 3) fabrication defects due to the limitations of our manufacturing facilities. Manufacturing these brakes is quite challenging due to the small parts with high tolerances. We manufactured several prototype brakes and continuously improved our ability to make better parts and align them well during assembly of the brakes. Still, slight misalignments could not be avoided and were considered to be the main reason for the fluctuations. In the haptic glove application, this was not critical. In a typical VR simulation, the user makes small finger movements to grasp an object. Hence, the brakes do not spin continuously as they would in other applications. The step response of a simulated first-order system was overlaid on the experimental torque data (thick line in Fig. 7), resulting in an estimated time constant of 67 ms for the small brake and 100 ms for the large brake.

Fig. 8 shows the torque versus input current characteristics. In the operational range of up to 0.5 A current for our prototypes, this relationship is fairly linear. In case of the small brake, magnetic saturation begins to take place ($B = 2.004$ T) in the steel flux path starting around 0.5 A current, which limits the torque output. The large brake starts to saturate around 0.7 A ($B = 2.03$ T). However, in both brakes, the coil wire gets hot once the current is more than 0.5 A. Table I provides design specifications of the final prototypes.

IV. MR GLOVE

Force feedback gloves need to: 1) provide sustained forces to multiple fingers; 2) be lightweight to minimize user fatigue; 3) be safe; and 4) preserve user's natural arm freedom of motion as much as possible. Keeping these general requirements in mind, we designed the MR glove (see Fig. 9) using the MR

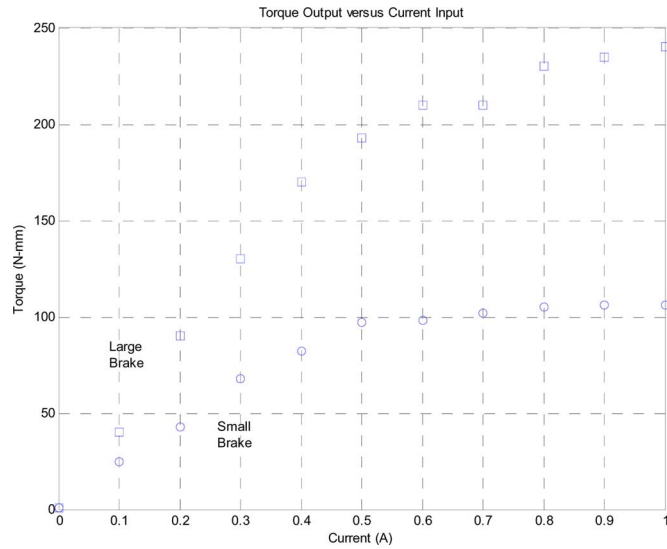


Fig. 8. Torque output versus current input for the large and small brakes.

TABLE I
DESIGN SPECIFICATIONS OF PROTOTYPES

	Small Brake	Large Brake
Spool width (mm)	8	13
Brake width* (mm)	21.5	26.5
Shell length (mm)	36.6	35.5
Shell height (mm)	28	28
Weight* (g)	68	84
Coil turns	150	250
MR-fluid volume (ml)	0.09	0.15
Torque output* (N-mm)	399	821
Time constant (ms)	67	100

*Values include 28:120 gear train.

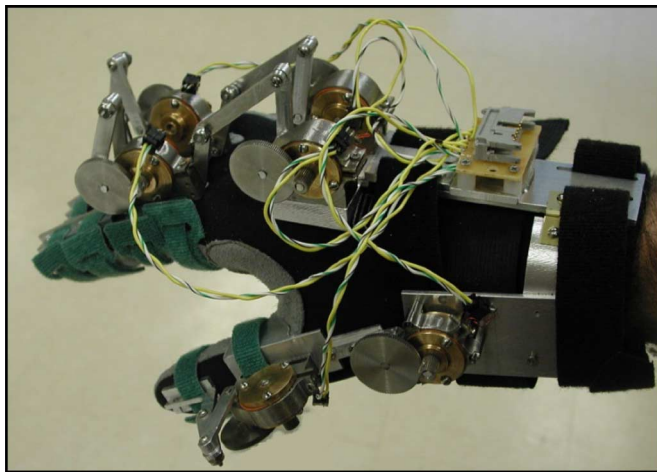


Fig. 9. MR glove.

brakes. It is worn over a commercially available CyberGlove that provides finger joint position data.

The space available on the back of the hand for actuator placement is quite limited. Therefore, some of the DOFs of the hand had to be constrained to limit the number of actuators required. The thumb, index, and middle fingers are actuated with

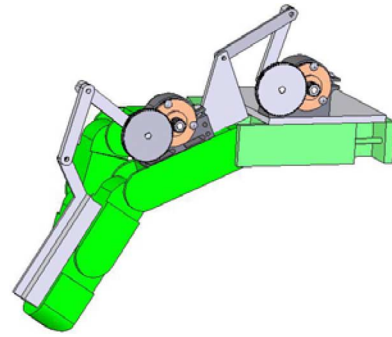


Fig. 10. Computer-aided design (CAD) model of a finger, brakes, and four-bar linkages.

two brakes on each. The thumb is constrained to 2 DOF. One small brake controls the distal joint while a large brake controls the motion of the proximal joint in the plane perpendicular to the plane of the hand. The distal (fingertip) and medial phalanges of the index finger are coupled together with a rigid link combining them into one phalange that pivots about the medial joint (see Fig. 10). This reduces the 3 DOF of the index finger into 2, namely, the medial joint (PIP) and the knuckle joint (MCP). The same constraints are also applied to the middle finger. Two large brakes are placed on the back of the hand to control the knuckle joints of the index and middle fingers. Two small brakes are placed on the proximal phalanges of the index and middle fingers to control the medial joints.

The braking torque is transmitted to each finger joint through the use of a four-bar linkage. This also allows the MR glove to be adjusted to different hand sizes. A four-bar linkage consists of ground, driver, coupler, and follower links. The index and middle fingers each have two 4-bar linkages and two brakes. The brakes are connected to the follower link of the four-bar linkages. When a brake is engaged, the rotation of the follower link is constricted, which then locks the driver link connected to a finger phalange, stopping the motion of that finger joint. The abduction motion of the thumb is constrained by a brake on the proximal joint. The distal phalange of the thumb is controlled by a small brake on the proximal phalange through a four-bar linkage. During virtual object manipulation, the users move their fingers slowly due to inherent delays in the simulation and to be able to feel the interaction forces with the objects. Therefore, the mechanisms were treated as quasi-static. Using standard techniques for four-bar mechanisms, both kinematic analysis and static force modeling were done [17].

The components of the MR glove were mounted on a diving glove made of durable neoprene material that fit snugly on the hand. The palm and the underside of the fingers were reinforced with synthetic leather material that provided a stiff base for the fastening straps. Velcro straps were sewn on the glove to firmly situate the brakes on the fingers.

Each brake on the glove was connected to a D/A channel of a data acquisition card (National Instruments PCI-6713) in the PC through a power transistor to regulate current through each brake. The VR simulation determined the amount of braking torque to be applied by each brake and set a proportional voltage

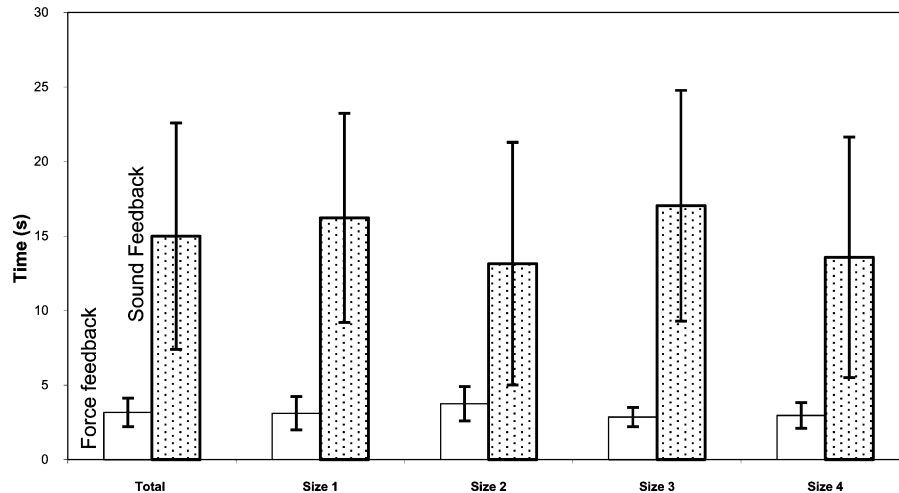


Fig. 11. Average task completion time to grasp a virtual object with force feedback (MR glove) and with sound (audio) feedback.

on each D/A channel. With this arrangement, each finger joint could be locked, simulating contact with a virtual surface, or a variable amount of torque could be applied to each joint, simulating various stiffness levels for a virtual object in hand.

The MR glove was connected to a dual-processor Pentium PC that also rendered the VR simulation. The virtual environment was created using the WorldToolKit (WTK) by Sense8 Company. It incorporated a CyberGlove by Immersion Company, two bird sensors by Ascension, Inc., and a ProView XL50 Head Mounted Display (HMD) by Kaiser Optoelectronics, Inc. The virtual hand model provided by the WTK for the CyberGlove has been used to map the user's real hand to the virtual world.

Since the design of the MR glove, two other haptic gloves using MRF have been reported in literature. The MRAGES uses a small linear cylinder-type actuator filled with MRF [25]. The actuator can produce up to 6-N force and has 32-mm stroke. It is connected to the fingertips using an innovative exoskeleton. The main drawback to the actuator design is that it produces 1.9-N force even in the OFF-state. This static force is about 32% of the actuator's maximum force range. It causes hand fatigue and may make this type of actuator more suited for stronger, less sensitive portions of the human body such as leg, arm, or ankle [25]. In comparison, the MR brake has only 5 N-mm static friction torque [T_{friction} in (2)], which is about 1.2% of the maximum torque output. Another glove, called smart glove, also uses a cylinder-type MRF actuator [26]. The actuator houses a Hall-effect sensor to measure the position of the piston. It weighs 52 g and has 20 mm stroke. A tendon-type flexible link was designed to connect the fingertips to the MRF actuators. The flexible link has a bow-shaped cross section, allowing it to bend at the finger joints yet be rigid elsewhere. In this design actuator, friction was stated to be a major concern to improve the transparency of the haptic interface.

V. USABILITY EXPERIMENTS

We conducted two groups of experiments to assess the effectiveness of the MR glove in: 1) virtual object manipulation and 2) reflecting stiffness of a virtual object held between two fingers. Fourteen users with no prior VR experience participated in

the experiments. After calibrating the CyberGlove and adjusting the MR glove to each user's hand, the user was allowed to work with the system for a while to get used to it.

A. Grasping Experiment

In this experiment, the user was asked to grasp a virtual cube from two opposing surfaces using his thumb and index finger. The goal was to determine if the MR glove had any effect in detecting contact with the virtual object. The hypothesis was that given the same set of virtual cubes, a user will be able to grasp the cubes quicker using the MR glove. We conducted the experiment with and without the MR glove activated. In the case of inactive MR glove, an audio feedback was provided to the user when he successfully grasped the object.

The user's back was turned to the computer screen so that he would not have any visual feedback from the VR simulation. His forearm was positioned on an armrest. First, the experiment administrator placed a virtual cube in between the thumb and index fingers of the virtual hand in the VR environment. Then, the user was asked to grasp the cube by closing his fingers. The task completion time was measured between the time the experiment started and the time successful grasp was achieved. In the case of the active MR glove experiments, the MR glove locked the brakes, providing a sensation of holding the cube between two fingers. In the case of the inactive MR glove experiments, the user still wore the MR glove. However, the only feedback he received was the audio feedback when both fingers came in contact with the opposing surfaces of the virtual cube and stayed in contact.

Four different sizes of virtual cubes were used in the experiment. The users did not know what size cube was placed in between their fingers. With the active MR glove, the average task completion time ($M = 3.16$ s, $SD = 0.95$) was much shorter than the case with the inactive glove ($M = 14.9$ s, $SD = 7.59$). The MR glove significantly improved the task completion times, $t(14) = 1.66$, $p < 0.05$. The experiments for a pick-and-place task showed that the glove reduced the task completion time by an average 79% (see Fig. 11). Although significant,

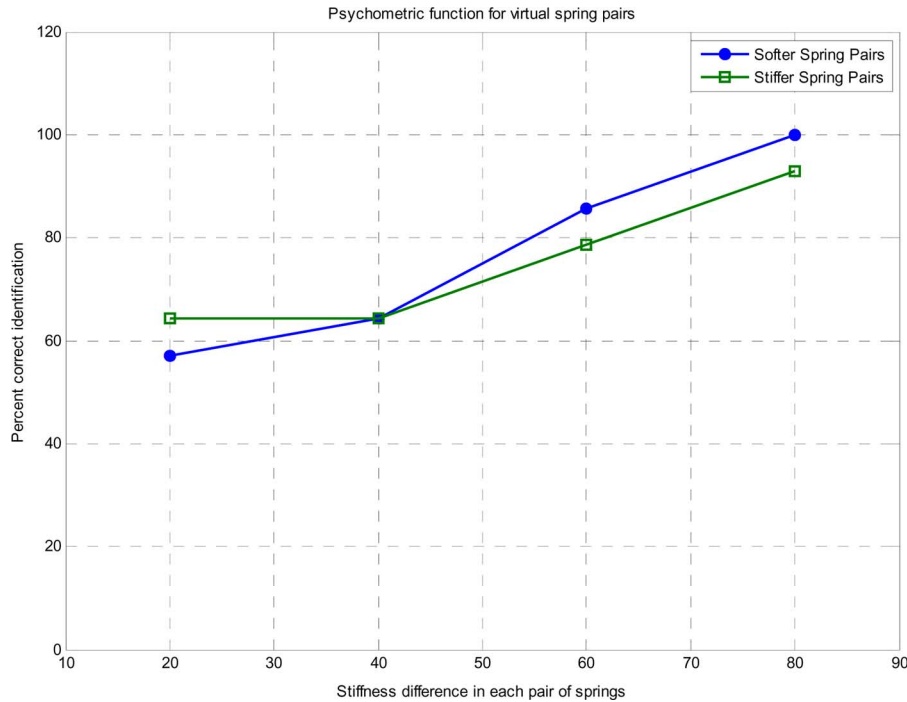


Fig. 12. Percentage of correct identifications in object stiffness experiments.

this was a somewhat expected result since many studies in literature showed that haptic interfaces improved task completion times. We observed a noticeable difference the grasping algorithm made in the realism of the simulations. Compared to conventional collision algorithms where the fingertip is used for collision detection, in our algorithm, small surface areas were defined on the underside of the finger digits for collision detection. This approach made grasping objects more realistic because it required the user to maintain surface contact and did not allow the fingers to penetrate into the object. Coupled with the near-rigid response from the MR brakes, simulation of rigid virtual objects was quite realistic. As mentioned earlier, slight misalignments could not be avoided due to the manufacturing challenges. The effect was torque fluctuations when the brake was spun by a motor. However, in its intended implementation on the MR glove, no such fluctuations or vibration could be felt by the users because the brakes did not spin at high speed. The fingers swept a limited range of motion and did so slowly while the users were trying to grasp the virtual objects.

B. Object Stiffness Experiment

In this experiment, the users squeezed a virtual spring between their thumb and index finger. The goal was to assess how well they could distinguish stiffness of a pair of virtual springs with the help of the MR glove. The hypothesis was that the larger the difference between the springs, the easier it would be to distinguish them. During the experiments, the thumb joint brakes were locked. The index finger started from fully open position and closed until it touched the thumb. This allowed the virtual spring between the fingers to be compressed always the same amount. This way, the users could not ascertain the virtual

spring stiffness by just looking at how much they could compress the spring. To project various stiffness values, the torque at the proximal joint of the index finger was varied. Therefore, the virtual spring stiffness was translated into braking torque at the index proximal joint. Fourteen users participated in the experiment. Users were presented with virtual springs A and B that looked identical on the screen. They were asked to identify the stiffer spring in the pair. The users were allowed to squeeze each spring as many times as they wanted.

Each user was given eight sets of virtual springs where each set consisted of springs A and B. The spring pairs were randomly selected from the available sets. Furthermore, whether spring A or B was the stiffer one was also randomized in each trial. The user was asked to identify the stiffer spring in each set. The spring pairs were characterized by the corresponding joint torques since the braking torque was controlled to present different stiffness values to the users. We used 20–40, 20–60, 20–80, and 20–100 N·mm as the joint torques representing springs A and B. It was expected that as the difference between the springs got smaller, the user would have greater difficulty identifying the stiffer spring. Furthermore, we expected that not only the relative stiffness values but also the absolute value of the spring stiffness would have an effect on the user's ability to discern the difference. For example, a 20–40 pair should be easier to distinguish than the 100–120 pair due to the 1:2 ratio in the softer spring set although the difference in each case is the same (20 N·mm). Therefore, the experiment was also conducted using stiffer set of springs with the pairs of 100–120, 100–140, 100–160, and 100–180 N·mm.

Fig. 12 shows that increased difference between the stiffness values in both the softer and the stiffer sets generates higher percentage of correct responses from the users. As indicated by

the two curves, the difference between the softer and stiffer sets is not significant. Hence, the absolute values of the stiffness did not seem to have an effect. In the cases of 60 and 80 N-mm differences, the users identified the virtual springs correctly with about 80% or higher success rate. For comparison, researchers reported 75% correct identification rate using the only commercially available force feedback glove called CyberGrasp [35]. In their experiments, subjects were asked to discern which of two virtual objects of similar size was stiffer. The stiffer object had about four times the stiffness of the softer one, which is similar to our case in the 20–80 spring set. In our experiments with 20 and 40 N-mm differences, the subjects tended to be less confident and took longer to make a decision. From a qualitative standpoint, these two cases seemed to be near the limit of perception. Since the probability of a correct answer by simply guessing is 50%, we concluded that in the cases of a difference of 20 and 40 N-mm, the users did not receive clear enough information from the MR glove to be able to distinguish the virtual springs.

VI. CONCLUSION

In this paper, a force feedback glove for VR has been presented. The haptic interface, called MR glove, provides the sensation of holding a virtual object in the VR world. When the digital hand model in the VR world grasps an object, the glove activates its brakes to stop the user's fingers, preventing his fingers from penetrating into the virtual object.

In the haptic glove applications, the challenge is to design an actuator that is small enough to be placed on the hand yet powerful enough to restrict or stop the motion of the fingers as the user grasps a virtual object.

In this research, we developed compact MR brakes and placed six of them on the glove. The brakes use two concentric cylinders. The inner cylinder (the drum) rotates inside the outer cylinder (the spool). The gap between the drum and the spool is filled with the MRF that applies a controllable braking torque on the drum due to the change in its viscosity as a function of the strength of the applied magnetic field. The primary goal was to design a brake that was about 25 mm in diameter and no wider than the width of a finger (about 15 mm) yet powerful enough to apply sufficient force on a finger joint to restrict or stop its free motion. We achieved this goal by stacking low-carbon steel rings and aluminum rings to create a serpentine flux path through the MRF to increase the volume of fluid exposed to the magnetic flux lines. By optimizing parameters such as the drum radius, width, fluid gap, number of coil turns, and current, we maximized the torque output. Two types of brakes with 821 and 399 N-mm torque were fabricated. They have 100 and 67 ms time constants and weigh 84 and 68 g. The static friction was experimentally determined as 5 N-mm.

We conducted two groups of experiments to assess the effectiveness of the MR glove in: 1) virtual object manipulation and 2) reflecting stiffness of a virtual object held between two fingers. In manipulation experiments, the user was asked to grasp a virtual cube from two opposing surfaces using his thumb and index finger. The goal was to determine if the MR glove had any effect in detecting contact with the virtual object. The MR glove

shortened the task completion times by 79%. In object stiffness experiments, the users squeezed a virtual spring between their thumb and index finger. The goal was to assess how well they could distinguish the stiffness of a pair of virtual springs with the help of the MR glove. It was expected that as the difference between the springs got smaller, the user would have greater difficulty identifying the stiffer spring. Furthermore, we expected that not only the relative stiffness values but also the absolute value of the spring stiffness would have an effect on the user's ability to discern the difference. Results confirmed that as the difference between the stiffness values of springs in a set increased, users were able to identify the stiffer spring with a higher percentage of success. However, the absolute values of the spring stiffness did not seem to make a difference.

The MR glove weighs 640 g. We are working on the next version of the glove to reduce its weight by using linkages made of composite materials. In addition, we are investigating hybrid designs using small motor and MR brake combinations, and closed-loop control implementations. The next-generation glove will also incorporate built-in position sensing capabilities to eliminate the need to wear a CyberGlove for finger position measurements.

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